

INTRODUCTION TO DESIGN PATTERNS

Behavioural Design Patterns
Creational Design Patterns
Structural Design Patterns

Design Patterns

A design pattern is a proven, **ready-to-use** solution for **common problems** that developers face repeatedly in software design.

Patterns are like **blueprints** or **templates** that you can adapt to solve similar problems in your code. They are **not ready-made code** that you copy-paste directly.

Design patterns focus on how **objects interact** with each other, how to **structure your system**, and how **classes/objects communicate** to solve problems efficiently.

Benefits of using Design Patterns

Reusable Solutions – Once you learn a pattern, you can apply it across different projects when facing similar problems. No need to reinvent the wheel.

Easier Maintenance – Patterns keep your code organized and clean, making it simpler to fix bugs or add features without breaking existing functionality.

Better Communication – When you say "Let's use the Singleton pattern," your team immediately understands. Patterns provide a common vocabulary for developers.

Future-Proof – Patterns help your code scale. When you need to handle more users or features, the structure adapts without major rewrites.

Saves Time – Use proven solutions tested by thousands of developers instead of experimenting from scratch.